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Semiconductor-on-insulator wafer having a composite insulator layer

Abstract

Various embodiments of the present disclosure are directed towards a semiconductor wafer. The semiconductor wafer comprises a handle wafer. A first oxide layer is disposed over the handle wafer. A device layer is disposed over the first oxide layer. A second oxide layer is disposed between the first oxide layer and the device layer, wherein the first oxide layer has a first etch rate for an etch process and the second oxide layer has a second etch rate for the etch process, and wherein the second etch rate is greater than the first etch rate.

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/192,333, filed on Mar. 4, 2021, which is a Divisional of U.S. application Ser. No. 16/580,259, filed on Sep. 24, 2019 (now U.S. Pat. No. 10,950,631, issued on Mar. 16, 2021). The contents of the above-referenced patent applications are hereby incorporated by reference in their entirety.

BACKGROUND

(1) Integrated circuits (ICs) have traditionally been formed on bulk semiconductor wafers. In recent years, semiconductor-on-insulator (SOI) wafers have emerged as an alternative to bulk semiconductor wafers. An SOI wafer comprises a handle wafer, a buried oxide layer overlying the handle wafer, and a device layer overlying the buried oxide layer. Among other things, an SOI wafer leads to reduced parasitic capacitance, reduced leakage current, reduced latch up, and improved semiconductor device performance (e.g., lower power consumption and higher switching speed).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. **1** illustrates a cross-sectional view of some embodiments of a semiconductor-on-insulator (SOI) wafer having a composite insulator layer.

(3) FIG. **2** illustrates a cross-sectional view of some other embodiments of the SOI wafer of FIG. **1**.

(4) FIG. **3** illustrates a cross-sectional view of some other embodiments of the SOI wafer of FIG. **1**.

(5) FIG. **4** illustrates a cross-sectional view of some embodiments of an integrated chip (IC) comprising a semiconductor-on-insulator (SOI) substrate having a composite insulator structure.

(6) FIG. **5** illustrates a cross-sectional view of some other embodiments of the IC of FIG. **4**.

(7) FIGS. **6-16** illustrate a series of cross-sectional views of some embodiments for forming a semiconductor-on-insulator (SOI) wafer having a composite insulator layer and singulating individual integrated chips (ICs) from the SOI wafer.

(8) FIG. **17** illustrates a flowchart of some embodiments of a method for forming a semiconductor-on-insulator (SOI) wafer having a composite insulator layer and singulating individual integrated chips (ICs) from the SOI wafer.

DETAILED DESCRIPTION

(9) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

(10) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship

to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(11) Some semiconductor-on-insulator (SOI) wafers comprise a handle wafer (e.g., a silicon wafer), a buried oxide layer (e.g., silicon dioxide (SiO.sub.2)) overlying the handle wafer, and a device layer (e.g., a silicon layer) overlying the insulator layer. One approach for forming such an SOI wafer is a bond and etch process. One of two bond and etch process are typically utilized to form the SOI wafer.

(12) One bond and etch process includes forming the oxide layer on the handle wafer via a thermal oxidation process. A device wafer is then bonded to the oxide layer. Thereafter, the device wafer is etched backed to form the device layer over the oxide layer. However, because the device wafer is bonded to the oxide layer, there is a bond interface between the device layer and the oxide layer. The bond interface is a source of leakage paths that negatively affect the performance of semiconductor devices (e.g., transistors) that are subsequently formed on the device layer.

(13) Another bond and etch process includes forming the oxide layer on the device wafer via the thermal oxidation process. The handle wafer is then bonded to the oxide layer. Thereafter, the device wafer is etched backed to form the device layer over the oxide layer. However, because the oxide layer is formed on the device wafer via the thermal oxidation process, the use of an etch stop layer for the etching back of the device layer is limited. For example, the thermal oxidation process prevents the use of an epitaxial etch stop layer due to the relatively high temperature (e.g., at least 1000° C.) of the thermal oxidation process relaxing the epitaxial etch stop layer. Because the thermal oxidation process prevents the use of an epitaxial etch stop layer, the total thickness variation (TTV) of the device layer is negatively affected.

(14) Various embodiments of the present application are directed toward a method for forming an SOI wafer. The method comprises forming an etch stop layer over a donor wafer. A device layer is formed over the etch stop layer. A first oxide layer is formed over the device layer via a chemical vapor deposition (CVD) process. A second oxide layer is formed over the handle wafer via a thermal oxidation process. The first oxide layer is then bonded to the second oxidation layer. Thereafter, the donor wafer and etch stop layer are removed to form the SOI wafer. Because the first oxide layer is formed by the CVD process, the first oxide layer may be formed on the device wafer without negatively affecting the use of the etch stop layer (e.g., due to the relatively low temperature (e.g., less than or equal to 900° C.) needed for the CVD process). Thus, the TTV of the device layer may be improved (e.g., reduced TTV). In addition, because the first oxide layer is formed on the device layer, a bond interface between the first oxide layer and the second oxide layer is disposed relatively far away from a bottom surface of the device layer. Because the bond interface is disposed relatively far away from the bottom surface of the device layer, the performance of integrated chips (e.g., dies) formed from the SOI wafer **100** may be improved due to a reduction in potential leakage paths.

(15) FIG. **1** illustrates a cross-sectional view of some embodiments of a semiconductor-on-insulator (SOI) wafer **100** having a composite insulator layer **104**.

(16) As shown in FIG. **1**, the SOI wafer **100** comprises a handle wafer **102**, a composite insulator layer **104**, and a device layer **106**. The SOI wafer **100** may be used with, for example, complementary metal-oxide-semiconductor (CMOS) applications, embedded flash applications, CMOS image sensor applications, near infrared (NIR) applications, microelectronics applications, optoelectronics applications, micro-electro-mechanicals systems (MEMS) applications, or the like. In some embodiments, the SOI wafer **100** has a circular top layout and/or has a diameter of approximately 200, 300, or 450 millimeters (mm). In other embodiments, the SOI wafer **100** may have some other shape and/or some other dimensions.

(17) The handle wafer **102** may be or comprise, for example, monocrystalline silicon, some other silicon material (e.g., polycrystalline silicon), some other semiconductor material (e.g., germanium (Ge)), or any combination of the foregoing. The device layer **106** overlies the handle wafer **102**. The device layer **106** is configured to be processed so that semiconductor devices (e.g., transistors) may be formed on the device layer **106**. The device layer **106** may be or comprise, for example, monocrystalline silicon, some other silicon material, some other semiconductor material, or any combination of the foregoing. In some embodiments, the device layer **106** may have a thickness (e.g., a distance between an upper surface and a lower surface) between 100 angstrom (Å) and 3000 Å. In further embodiments, the thickness of the device layer **106** may be 1300 Å. In yet further embodiments, the device layer **106** may be an epitaxial layer (e.g., formed by an epitaxy process).
(18) The composite insulator layer **104** is disposed between the handle wafer **102** and the device layer **106**. The composite insulator layer **104** comprises a first insulator layer **108** and a second insulator layer **110**. The first insulator layer **108** contacts the second insulator layer **110** at a bond interface **112**, such that material of the first insulator layer **108** is bonded to the material of the second insulator layer **110**. In some embodiments, the bond interface **112** comprises dielectric-to-dielectric bonds between the material of the first insulator layer **108** and the material of the second insulator layer **110**. In further embodiments, the first insulator layer **108** contacts the handle wafer **102**. In yet further embodiments, the second insulator layer **110** contacts the device layer **106**.
(19) In some embodiments, the composite insulator layer **104** may have a thickness between 200 Å and 2 micrometers (um). The first insulator layer **108** may have a thickness between 100 Å and 1 um. The second insulator layer **110** may have a thickness between 100 Å and 1 um. In further embodiments, a ratio of the thickness of the second insulator layer **110** to the thickness of the first insulator layer **108** is between 0.1 and 10.
(20) The first insulator layer **108** may be or comprise, for example, an oxide (e.g., SiO.sub.2), a high-k dielectric (e.g., a dielectric material with a dielectric constant greater than 3.9), or the like. In embodiments in which the first insulator layer **108** is an oxide (e.g., SiO2), the first insulator layer **108** may be referred to as a first oxide layer. In further embodiments, the first insulator layer **108** may be a thermal oxidation oxide. For example, the thermal oxidation oxide may be silicon dioxide formed by a thermal oxidation process.
(21) The second insulator layer **110** may be or comprise, for example, an oxide (e.g., SiO.sub.2), a high-k dielectric (e.g., a dielectric material with a dielectric constant greater than 3.9), or the like. In embodiments in which the second insulator layer **110** is an oxide (e.g., SiO2), the second insulator layer **110** may be referred to as a second oxide layer. In further embodiments, the second insulator layer **110** may be a chemical vapor deposition (CVD) oxide. For example, the CVD oxide may be silicon dioxide formed by a CVD process, such as, plasma-enhanced CVD (PECVD), low pressure CVD (LPCVD), high-density plasma CVD (HDPCVD), or the like. Because the SOI wafer **100** comprises the composite insulator layer **104**, the bond interface **112** is disposed a relatively large distance from a bottom surface of the device layer **106**. Because the bond interface **112** is disposed relatively far away from the bottom surface of the device layer **106**, the performance of integrated chips (e.g., dies) formed from the SOI wafer **100** may be improved due to a reduction in potential leakage paths.
(22) FIG. **2** illustrates a cross-sectional view of some other embodiments of the SOI wafer **100** of FIG. **1**.
(23) As shown in FIG. **2**, the first insulator layer **108** surrounds the handle wafer **102**. In such embodiments, the first insulator layer **108** may extend continuously along an upper surface of the handle wafer **102**, along a first sidewall of the handle wafer **102**, along a bottom surface of the handle wafer **102**, and along a second sidewall of the handle wafer **102** opposite the first sidewall. In further such embodiments, the composite insulator layer **104** comprises the second insulator layer **110** and a portion of the first insulator layer **108** disposed between the handle wafer **102** and

the device layer **106**.

(24) In some embodiments, the second insulator layer **110** has a higher concentration of a predefined chemical element than the first insulator layer **108**. The predefined chemical element may be, for example, hydrogen (H), carbon (C), chlorine (Cl), or the like. In further embodiments, a thermal stability of the second insulator layer **110** at a predefined temperature (e.g., between 600° C. and 1200° C.) may be the same as a thermal stability of the first insulator layer **108** at the predefined temperature. For example, the second insulator layer **110** and the first insulator layer **108** may be stable at the predefined temperature. In other embodiments, the thermal stability of the second insulator layer **110** may be different than the thermal stability of the first insulator layer **108**. For example, the first insulator layer **108** may be stable at the predefined temperature and the second insulator layer **110** may densify (e.g., become denser) at the predefined temperature, or the first insulator layer **108** may be stable at the predefined temperature and the second insulator layer **110** may eject (e.g., outgas) some of the predefined chemical at the predefined temperature.

(25) In some embodiments, a density of the second insulator layer **110** is between 2.1 gram per cubic centimeter (g/cm.sup.3) and 2.3 g/cm.sup.3. In further embodiments, the density of the second insulator layer **110** and a density of the first insulator layer **108** may be the same. For example, both the density of the first insulator layer **108** and the density of the second insulator layer **110** may be 2.2 g/cm.sup.3. In other embodiments, the density of the second insulator layer **110** may be different than the density of the first insulator layer **108**. For example, the density of the second insulator layer **110** may be greater than the density of the first insulator layer **108** (e.g., 2.3 g/cm.sup.3 and 2.2 g/cm.sup.3, respectively), or the density of the second insulator layer **110** may be less than the density of the first insulator layer **108** (e.g., 2.1 g/cm.sup.3 and 2.2 g/cm.sup.3, respectively).

(26) In some embodiments, an intrinsic stress of the second insulator layer **110** is between 3 megapascal (MPa) compressive and 3 MPa tensile. In further embodiments, the intrinsic stress of the second insulator layer **110** may be 3 MPa tensile, 1 MPa compressive, or 3 MPa compressive. In yet further embodiments, the intrinsic stress of the second insulator layer **110** may be the same as an intrinsic stress of the first insulator layer **108**. For example, both the intrinsic stress of the first insulator layer **108** and the intrinsic stress of the second insulator layer **110** may be 3 MPa compressive. In other embodiments, the intrinsic stress of the second insulator layer **110** may be different than the intrinsic stress of the first insulator layer **108**. For example, the intrinsic stress of the first insulator layer **108** may be 3 MPa compressive and the intrinsic stress of the second insulator layer may be 2 MPa compressive, 1 MPa compressive, 1 MPa tensile, 2 MPa tensile, 3 MPa tensile, or some other intrinsic stress value that is different than the intrinsic stress of the first insulator layer **108**.

(27) In some embodiments, a dielectric strength of the second insulator layer **110** is less than 11 megavolt per centimeter (MV/cm). In further embodiments, a dielectric strength of the first insulator layer **108** is greater than or equal to 11 MV/cm. In further embodiments, the dielectric strength of the second insulator layer **110** is between 5 MV/cm and 10 MV/cm. In yet further embodiments, the dielectric strength of the second insulator layer **110** may be 5 MV/cm, 8 MV/cm, or 10 MV/cm. The dielectric strength of the second insulator layer **110** may be different than a dielectric strength of the first insulator layer **108**. For example, the dielectric strength of the first insulator layer **108** may be 11 MV/cm and the dielectric strength of the second insulator layer **110** may be 5 MV/cm, 8 MV/cm, or 10 MV/cm.

(28) In some embodiments, the first insulator layer **108** has a first etch rate for a predefined etching process, and the second insulator layer **110** has a second etch rate for the predefined etching process that is different than the first etch rate. In further embodiments, the first etch rate is less than the second etch rate. The first etch rate may be less than or equal to 25 angstroms per minute (Å/min). The second etch rate may be between 400 Å/min and 30 Å/min. In yet further embodiments, the second etch rate is 400 Å/min, 60 Å/min, or 30 Å/min. In further embodiments,

the predefined etching process is a hydrofluoric (HF) etching process (e.g., HF acid etching process). In yet further embodiments, the HF etching process utilizes a HF acid solution having a ratio of water (H_2O) to HF acid of 100:1.
(29) In some embodiments, the first insulator layer **108** is a conformal layer that conforms to the contours of the handle wafer **102**. In further embodiments, the second insulator layer **110** is a conformal layer that conforms to the contours of the bottom surface of the device layer **106**. In other embodiments, the second insulator layer **110** is a non-conformal layer. In yet further embodiments, the first insulator layer **108** is a conformal layer and the second insulator layer **110** is a conformal layer. In other embodiments, the second insulator layer **110** is a non-conformal layer and the first insulator layer **108** is a conformal layer.
(30) FIG. **3** illustrates a cross-sectional view of some other embodiments of the SOI wafer **100** of FIG. **1**.
(31) In some embodiments, outermost sidewalls of the second insulator layer **110** are disposed between outermost sidewalls of the first insulator layer **108**, such that an edge region of the SOI wafer **100** has a step-like profile. The outermost sidewalls of the second insulator layer **110** may be disposed between outermost sidewalls of the handle wafer **102**. In further embodiments, outermost sidewalls of the device layer **106** are disposed between the outermost sidewalls of the first insulator layer **108**. The outermost sidewalls of the device layer **106** may be disposed between the outermost sidewalls of the handle wafer **102**. In further embodiments, the outermost sidewalls of the device layer **106** are substantially aligned with the outermost sidewalls of the second insulator layer **110**. In yet further embodiments, the outermost sidewall of the device layer **106** and/or the outermost sidewalls of the second insulator layer **110** may extend vertically at an angle that is substantially perpendicular to an upper surface of the first insulator layer **108**. In other embodiments, the outermost sidewall of the device layer **106** and/or the outermost sidewalls of the second insulator layer **110** may be angled (e.g., angled inward or outward).
(32) In some embodiments, the outermost sidewalls of the first insulator layer **108** may be disposed between the outermost sidewalls of the device layer **106** and/or the outermost sidewalls of the handle wafer **102**. In further embodiments, the outermost sidewalls of the second insulator layer **110** may be disposed between the outermost sidewalls of the device layer **106** and/or the outermost sidewalls of the handle wafer **102**. In yet further embodiments, the outermost sidewalls of the second insulator layer **110** may be disposed between the outermost sidewalls of the first insulator layer **108**.
(33) FIG. **4** illustrates a cross-sectional view of some embodiments of an integrated chip (IC) **400** comprising a semiconductor-on-insulator (SOI) substrate **401** having a composite insulator structure **404**.
(34) As shown in FIG. **4**, the IC **400** comprises an SOI substrate **401**. The SOI substrate **401** is a portion of the SOI wafer **100**. The SOI substrate **401** comprises a handle substrate **402**, a composite insulator structure **404**, and a device substrate **406**. The handle substrate **402** is a portion of the handle wafer **102**. The device substrate **406** is a portion of the device layer **106**. The composite insulator structure **404** is a portion of the composite insulator layer **104**.
(35) The composite insulator structure **404** comprises a first insulator structure **408** and a second insulator structure **410**. The first insulator structure **408** is a portion of the first insulator layer **108**. The second insulator structure **410** is a first portion of the second insulator layer **110**. The first insulator structure **408** contacts the second insulator structure **410** at the bond interface **112**, such that material of the first insulator structure **408** is bonded to the material of the second insulator structure **410**.
(36) One or more semiconductor devices **412** are disposed on/over the device substrate **406**. The one or more semiconductor devices **412** may be or comprise, for example, metal-oxide-semiconductor (MOS) field-effect transistors (FETs), some other MOS devices, or some other semiconductor devices. In some embodiments, each of the one or more semiconductor devices **412**

comprises a pair of source/drain regions **414**, a gate dielectric **416**, and a gate electrode **418**. In further embodiments, one or more isolation structures **420** (e.g., shallow trench isolation (STI) structures) are disposed in the device substrate **406**. The one or more isolation structures **420** may laterally surround the one or more semiconductor devices **412**. In yet further embodiments, the one or more isolation structures **420** may extend through the device substrate **406** to contact the second insulator structure **410**. In other embodiments, the one or more isolation structures **420** may be vertically spaced from the second insulator structure **410**.
(37) An interlayer dielectric (ILD) layer **422** is disposed over the device substrate **406** and the one or more semiconductor devices **412**. The ILD layer **422** may comprise, for example, an oxide (e.g., SiO.sub.2), a low-k dielectric (e.g., a dielectric material with a dielectric constant less than about 3.9), or the like. A plurality of conductive contacts **424** (e.g., tungsten contacts) are disposed in the ILD layer **422**. In some embodiments, the plurality of conductive contacts **424** extend through the ILD layer **422** to the source/drain regions **414** and/or the gate electrode **418** of each of the one or more semiconductor devices **412**.
(38) Although not shown, additional dielectric layers and conductive features may be disposed over the ILD layer **422** and the conductive contacts **424**. For example, one or more additional ILD layers, conductive wires (e.g., copper wires), conductive vias (e.g., copper vias), and/or passivation layers may be disposed over the ILD layer **422**. In such embodiments, the ILD layers may be collectively referred to as an ILD structure, and the conductive features may collectively be referred to as an interconnect structure (e.g., copper interconnect structure).
(39) In some embodiments, outermost sidewalls of the ILD layer **422** are substantially aligned with outermost sidewalls of the device substrate **406**. The outermost sidewalls of the device substrate **406** may be substantially aligned with outermost sidewalls of the second insulator structure **410**. In further embodiments, the outermost sidewalls of the second insulator structure **410** are substantially aligned with outermost sidewalls of the first insulator structure **408**. The outermost sidewalls of the first insulator structure **408** may be substantially aligned with outermost sidewalls of the handle substrate **402**.
(40) In some embodiments, the outermost sidewalls of the first insulator structure **408** may be disposed between the outermost sidewalls of the device substrate **406** and/or the outermost sidewalls of the handle substrate **402**. In further embodiments, the outermost sidewalls of the second insulator structure **410** may be disposed between the outermost sidewalls of the device substrate **406** and/or the outermost sidewalls of the handle substrate **402**. In yet further embodiments, the outermost sidewalls of the second insulator structure **410** may be disposed between the outermost sidewalls of the first insulator structure **408**.
(41) FIG. **5** illustrates a cross-sectional view of some other embodiments of the IC **400** of FIG. **4**.
(42) As shown in FIG. **5**, the IC **400** may comprise a third insulator structure **502**. In some embodiments, the third insulator structure **502** is a second portion of the first insulator layer **108**. The third insulator structure **502** may have a same chemical composition as the first insulator structure **408**. In further embodiments, the third insulator structure **502** has a third etch rate for the predefined etching process that is the same as the first etch rate. The third insulator structure **502** and the first insulator structure **408** may have a same thermal stability, density, intrinsic stress, and/or dielectric strength. In further embodiments, the third insulator structure **502** conforms to the contours of a bottom surface of the handle substrate **402**. In yet further embodiments, outermost sidewalls of the third insulator structure **502** may be substantially aligned with the outermost sidewalls of the handle substrate **402**.
(43) FIGS. **6-16** illustrate a series of cross-sectional views of some embodiments for forming a semiconductor-on-insulator (SOI) wafer **100** having a composite insulator layer **104** and singulating individual integrated chips (ICs) from the SOI wafer **100**.
(44) As shown in FIG. **6**, a first insulator layer **108** is formed on a handle wafer **102**. In some embodiments, the first insulator layer **108** is formed on an upper surface of the handle wafer **102**.

In further embodiments, the first insulator layer **108** is formed as a continuous layer on the upper surface of the handle wafer **102**, a first sidewall of the handle wafer **102**, a bottom surface of the handle wafer **102**, and a second sidewall of the handle wafer **102** opposite the first sidewall. In yet further embodiments, the first insulator layer **108** is formed as a conformal layer.

(45) In some embodiments, a process for forming the first insulator layer **108** comprises growing the first insulator layer **108** via a thermal oxidation process. In further embodiments, the thermal oxidation process comprises oxidizing the handle wafer **102** in a processing chamber. In yet further embodiments, the thermal oxidation process comprises loading the handle wafer into the processing chamber, heating the handle wafer to a first processing temperature, and flowing a processing fluid into the processing chamber. The first processing temperature may be greater than or equal to 800° C. In further embodiments, the first processing temperature may be greater than or equal to 1000° C. The processing fluid may comprise, for example, oxygen (O), hydrogen (H), a combination of the foregoing, or some other processing fluid suitable for oxidizing the handle wafer **102**.

(46) In some embodiments, a planarization process (e.g., chemical-mechanical polishing (CMP)) may be performed on the handle wafer **102** and/or the first insulator layer **108** to reduce a thickness of the handle wafer **102**. The thickness of the handle wafer **102** may be reduced to less than or equal to 2 um. In further embodiments, the thickness of the handle wafer is reduced to 1.9 um.

(47) As shown in FIG. **7**, a processing layer **704** is formed over a donor wafer **702**. In some embodiments, the processing layer **704** is formed on the donor wafer **702**. The donor wafer **702** may comprise any type of semiconductor body (e.g., monocrystalline silicon/CMOS bulk, silicon-germanium (SiGe), silicon on insulator (SOI), etc.). In some embodiments, the donor wafer **702** is doped with first doping type dopants (e.g., p-type dopants). In further embodiments, the donor wafer **702** has a first doping concentration of the first doping type dopants.

(48) In some embodiments, the processing layer **704** is a semiconductor (e.g., silicon, germanium, etc.). In such embodiments, the processing layer **704** may be referred to as a semiconductor layer. In further embodiments, the processing layer **704** is silicon (e.g., monocrystalline silicon, polycrystalline silicon, etc.). The processing layer **704** may be doped with the first doping type dopants. The processing layer **704** may have a second doping concentration of the first doping type dopants that is less than the first doping concentration.

(49) In some embodiments, the processing layer **704** is an epitaxial layer (e.g., formed by an epitaxy process). In further embodiments, the processing layer **704** may have a thickness less than or equal to 2 um. In other embodiments, the processing layer **704** may have a thickness greater than 2 um. In further embodiments, the thickness of the processing layer **704** may be 1.8 um. In yet further embodiments, a process for forming the processing layer **704** comprises depositing or growing the processing layer **704** by, for example, a CVD process, an epitaxy process, or the like.

(50) Also shown in FIG. **7**, an etch stop layer **706** is formed over the processing layer **704**. In some embodiments, the etch stop layer **706** is formed on the processing layer **704**. The etch stop layer **706** may comprise, for example, silicon (Si), germanium (Ge), oxygen (O), boron (B), arsenic (As), or the like. In some embodiments, the etch stop layer **706** is an epitaxial etch stop layer (e.g., formed by an epitaxy process).

(51) In some embodiments, the etch stop layer **706** may have a thickness less than or equal to 20 nanometers (nm). In other embodiments, the thickness of the etch stop layer **706** may be greater than 20 nm. In further embodiments, the thickness of the etch stop layer **706** may be 15 nm. In further embodiments, a process for forming the etch stop layer **706** comprises depositing or growing the processing layer **704** by, for example, a CVD process, an epitaxy process, or the like.

(52) Also shown in FIG. **7**, a device layer **106** is formed over the etch stop layer **706**. In some embodiments, the device layer **106** is formed on the etch stop layer **706**. The device layer **106** may be an epitaxial layer (e.g., formed by an epitaxy process). In further embodiments, the device layer **106**, the etch stop layer **706**, and the processing layer **704** are each an epitaxial layer. In yet further

embodiments, a process for forming the device layer **106** comprises depositing or growing the device layer **106** by, for example, a CVD process, an epitaxy process, or the like.
(53) In some embodiments, the etch stop layer **706** may be an implant doped etch stop layer. In such embodiments, the etch stop layer **706** may comprise the first doping type dopants or second doping type dopants (e.g., n-type dopants). In further such embodiments, the etch stop layer **706** may be disposed in the donor wafer **702**, the processing layer **704**, and/or the device layer **106**. In yet further embodiments, the donor wafer **702**, the processing layer **704**, the etch stop layer **706**, and the device layer **106** may be collectively referred to as a workpiece **708**.
(54) As shown in FIG. **8**, a second insulator layer **110** is formed over the device layer **106**. In some embodiments, the second insulator layer **110** is formed on the device layer **106**. In further embodiments, the second insulator layer **110** is formed as a conformal layer. In other embodiments, the second insulator layer **110** is formed as a non-conformal layer. In yet further embodiments, a process for forming the second insulator layer **110** comprises depositing the second insulator layer **110** via a CVD process. For example, the second insulator layer **110** may be deposited by PECVD, LPCVD, HDPCVD, or the like.
(55) In some embodiments, the second insulator layer **110** may be formed by a first PECVD. In further embodiments, the first PECVD process comprises forming the second insulator layer **110** over the device layer **106** in a processing chamber. In further embodiments, the first PECVD process comprises loading the workpiece **708** into the processing chamber, heating the workpiece **708** to a second processing temperature, and flowing one or more processing fluids into the processing chamber. The second processing temperature may be less than 800° C. In further embodiments, the second processing temperature is less than or equal to 200° C. The one or more processing fluids may be or comprise, for example, silane (SiH.sub.4), oxygen (O), or the like. In yet further embodiments, the first PECVD process may form the second insulator layer **110** as a non-conformal layer.
(56) In some embodiments, the second insulator layer **110** may be formed by a first LPCVD. In further embodiments, the first LPCVD process comprises forming the second insulator layer **110** over the device layer **106** in a processing chamber. In further embodiments, the first LPCVD process comprises loading the workpiece **708** into the processing chamber, heating the workpiece **708** to a third processing temperature, and flowing one or more processing fluids into the processing chamber. The third processing temperature may be less than 800° C. In further embodiments, the third processing temperature is less than or equal to 450° C. The one or more processing fluids may be or comprise, for example, SiH.sub.4, oxygen (O), or the like. In yet further embodiments, the first LPCVD process may form the second insulator layer **110** as a non-conformal layer.
(57) In some embodiments, the second insulator layer **110** may be formed by a second LPCVD. In further embodiments, the second LPCVD process comprises forming the second insulator layer **110** over the device layer **106** in a processing chamber. In further embodiments, the second LPCVD process comprises loading the workpiece **708** into the processing chamber, heating the workpiece **708** to a fourth processing temperature, and flowing one or more processing fluids into the processing chamber. The fourth processing temperature may be less than 800° C. In further embodiments, the fourth processing temperature may be less than or equal to 700° C. The one or more processing fluids may be or comprise, for example, carbon (C), oxygen (O), hydrogen (H), tetraethyl orthosilicate (TEOS), or the like. In yet further embodiments, the second LPCVD process may form the second insulator layer **110** as a conformal layer.
(58) In some embodiments, the second insulator layer **110** may be formed by a third LPCVD. In further embodiments, the third LPCVD process comprises forming the second insulator layer **110** over the device layer **106** in a processing chamber. In further embodiments, the third LPCVD process comprises loading the workpiece **708** into the processing chamber, heating the workpiece **708** to a fifth processing temperature, and flowing one or more processing fluids into the

processing chamber. The fifth processing temperature may be less than 1000° C. In further embodiments, the fifth processing temperature may be less than or equal to 900° C. The one or more processing fluids may be or comprise, for example, oxygen (O), nitrogen (N), hydrogen (H), chlorine (Cl), dichlorosilane (SiCl.sub.2H.sub.2), or the like. In yet further embodiments, the third LPCVD process may form the second insulator layer **110** as a conformal layer. While several examples of CVD processes for forming the second insulator layer **110** are provided above, it will be appreciated that, in some embodiments, other CVD process having their own specific operating conditions (e.g., processing temperatures, processing fluids, etc.) may be utilized to form the second insulator layer **110**.

(59) As shown in FIG. **9**, in some embodiments, the second insulator layer **110**, the device layer **106**, the etch stop layer **706**, the processing layer **704**, and the donor wafer **702** are patterned to remove edge regions of the second insulator layer **110**, the device layer **106**, the etch stop layer **706**, the processing layer **704**, and the donor wafer **702**, respectively. In some embodiments, the patterning comprises forming a patterned masking layer (not shown) (e.g., negative/positive photoresist) on the device layer **106**. Thereafter, the second insulator layer **110**, the device layer **106**, the etch stop layer **706**, the processing layer **704**, and the donor wafer **702** are exposed to an etchant (e.g., wet/dry etchant) to remove unmasked portions of the second insulator layer **110**, the device layer **106**, the etch stop layer **706**, the processing layer **704**, and the donor wafer **702**. Subsequently, in some embodiments, the patterned masking layer is stripped away. It will be appreciated that, in some embodiments, multiple patterned masking layers and/or multiple etchants may be utilized to remove the edge regions of the second insulator layer **110**, the device layer **106**, the etch stop layer **706**, the processing layer **704**, and the donor wafer **702**.

(60) As shown in FIG. **10**, the second insulator layer **110** is bonded to the first insulator layer **108**, thereby forming a composite insulator layer **104** comprising the first insulator layer **108** and the second insulator layer **110**. In some embodiments, the second insulator layer **110** is bonded to the first insulator layer **108** by, for example, direct bonding, vacuum bonding, or the like. By bonding the second insulator layer **110** to the first insulator layer **108**, a bond interface **112** is formed between the second insulator layer **110** and the first insulator layer **108**. In further embodiments, the bond interface **112** comprises dielectric-to-dielectric bonds between a material of the first insulator layer **108** and a material of the second insulator layer **110**.

(61) As shown in FIG. **11**, the donor wafer **702** (see, e.g., FIG. **7**) is removed from the processing layer **704**. In some embodiments, a process for removing the donor wafer **702** from the processing layer **704** comprises a first etching process that comprises exposing the donor wafer **702** to a first etchant (e.g., wet/dry etchant). In further embodiments, the process for removing the donor wafer **702** from the processing layer **704** comprises performing a grinding process on the donor wafer. A planarization process (e.g., CMP) may then be performed on the donor wafer **702**. Thereafter, the first etching process may be performed on the donor wafer **702**. In yet further embodiments, the first etchant may comprise, for example, hydrogen (H), fluorine (F), oxygen (O), carbon (C), nitrogen (N), or the like.

(62) As shown in FIG. **12**, the processing layer **704** (see, e.g., FIG. **7**) is removed from the etch stop layer **706**. In some embodiments, a process for removing the processing layer **704** from the etch stop layer **706** comprises a second etching process that comprises exposing the processing layer **704** to a second etchant (e.g., wet/dry etchant). In further embodiments, the process for removing the processing layer **704** from the etch stop layer **706** comprises performing a planarization process (e.g., CMP) on the processing layer **704**. Thereafter, the second etching process may be performed on the processing layer **704**. The etch stop layer **706** is less selective to the second etchant than the processing layer **704**, thereby terminating the second etching process at the etch stop layer **706**. In further embodiments, the second etchant may comprise, for example, hydrogen (H), oxygen (O), carbon (C), nitrogen (N), or the like. In yet further embodiments, the second etchant may be different than the first etchant.

(63) As shown in FIG. **13**, the etch stop layer **706** (see, e.g., FIG. **7**) is removed from the device layer **106**. In some embodiments, a process for removing the etch stop layer **706** from the device layer **106** comprises a third etching process that comprises exposing the etch stop layer to a third etchant (e.g., wet/dry etchant). In further embodiments, the third etchant may comprise, for example, hydrogen (H), fluorine (F), oxygen (O), carbon (C), nitrogen (N), or the like. The third etchant is different than the second etchant. In yet further embodiments, after the etch stop layer is removed, formation of the SOI wafer **100** (see, e.g., FIG. **1**) is complete.
(64) Because the second insulator layer **110** is formed by a CVD process, the processing temperature to form the second insulator layer **110** may be relatively low (e.g., less than or equal to 900° C., 800° C., 700° C., 450° C., or 200° C.). Because the processing temperature to form the second insulator layer **110** is relatively low, the second insulator layer **110** may be formed on the device layer **106** without negatively affecting the etch stop layer **706**. Thus, the total thickness variation (TTV) of the device layer **106** may be improved (e.g., a reduction in TTV).
(65) For example, if the etch stop layer **706** is an epitaxial layer, the relatively low temperature may not undesirably relax the etch stop layer **706**. Because the relatively low temperature may not undesirably relax the etch stop layer **706**, after the second etching process, a TTV of the etch stop layer **706** may be improved. In addition, if the etch stop layer **706** is an implant doped etch stop layer, the relatively low temperature may widen the doping profile of the etch stop layer **706**. Because the relatively low temperature may widen the doping profile of the etch stop layer **706**, after the second etching process, a TTV of the etch stop layer **706** may be improved. Because the TTV of the etch stop layer **706** may be improved after the second etching process, the TTV of the device layer **106** may be improved (e.g., due to the improved TTV of the etch stop layer translating into an improved TTV of the device layer after the third etching process).
(66) In some embodiments, after the etch stop layer **706** is removed from the device layer **106**, the device layer **106** may be thinned down. In some embodiments, the device layer may be thinned down to a thickness between 100 Å and 3000 Å. In further embodiments, the device layer **106** may be thinned down by a thinning process, for example, an anneal process, a baking process, a planarization process (e.g., CMP), some other thinning process, or a combination of the foregoing.
(67) As shown in FIG. **14**, one or more isolation structures **420** (e.g., shallow trench isolation (STI) structures) are formed in the device layer **106**. In some embodiments, a process for forming the one or more isolation structures **420** comprises forming a patterned masking layer (not shown) on the device layer **106**. The device layer **106** is then exposed to an etchant to remove unmasked portions of the device layer **106**, thereby forming one or more trenches in the device layer **106**. Subsequently, in some embodiments, the patterned masking layer is stripped away. Thereafter, a dielectric layer (not shown) is deposited or grown on the device layer **106** and in the one or more trenches. A planarization process (e.g., CMP) is then performed on the dielectric layer, thereby forming the one or more isolation structures **420**. In further embodiments, the one or more isolation structures **420** may be formed extending through the device layer **106** to the second insulator layer **110**. In other embodiments, the one or more isolation structures **420** may be formed in the device layer **106** so that the one or more isolation structures **420** are vertically spaced from the second insulator layer **110**.
(68) Also shown in FIG. **14**, one or more semiconductor devices **412** are formed on/over the device layer **106**. In some embodiments, a process for forming the one or more semiconductor devices **412** comprises depositing or growing a gate dielectric layer (not shown) (e.g., SiO.sub.2) on the device layer **106**. A gate electrode layer (not shown) (e.g., polysilicon) is then deposited on the gate dielectric layer. A patterned masking layer (not shown) is then formed on the gate electrode layer. The gate electrode layer and the gate dielectric layer are then exposed to an etchant to remove unmasked portions of gate electrode layer and the gate dielectric layer, thereby forming a gate electrode **418** and a gate dielectric **416** for each of the one or more semiconductor devices **412**. Subsequently, in some embodiments, the patterned masking layer is stripped away. Thereafter,

source/drain regions **414** are formed in the device layer **106**. In some embodiments, the source/drain regions **414** may be formed by an ion implantation process (e.g., a self-aligned ion implantation process). In further embodiments, the above layers and/or structures may be deposited or grown by, for example, CVD, physical vapor deposition (PVD), atomic layer deposition (ALD), thermal oxidation, sputtering, some other deposition or growth process, or a combination of the foregoing.

(69) As shown in FIG. **15**, an interlayer dielectric (ILD) layer **422** is formed over the one or more semiconductor devices **412**, the device layer **106**, and the first insulator layer **108**. In some embodiments, the ILD layer **422** is formed contacting the device layer **106**, the one or more isolation structures **420**, the second insulator layer **110**, and the first insulator layer **108**. In further embodiments, a process for forming the ILD layer **422** comprises depositing the ILD layer **422** by, for example, CVD, PVD, sputtering, or the like. In yet further embodiments, a planarization process (e.g., CMP) may be performed on the ILD layer **422** to planarize an upper surface of the ILD layer **422**.

(70) Also shown in FIG. **15**, a plurality of conductive contacts **424** are formed extending through the ILD layer **422** to the source/drain regions **414** and/or the gate electrode **418** of each of the one or more semiconductor devices **412**. In some embodiments, a process for forming the plurality of conductive contacts **424** comprises forming a patterned masking layer on the ILD layer **422**. Thereafter, the ILD layer **422** is exposed to an etchant to remove unmasked portions of ILD layer **422**, thereby forming a plurality of conductive contact openings in the ILD layer **422**. Subsequently, in some embodiments, the patterned masking layer is stripped away. A conductive material (e.g., tungsten) is then deposited on the ILD layer **422** and in the plurality of conductive contact openings. The conductive material may be deposited by, for example, CVD, PVD, ALD, sputtering, electrochemical plating, electroless plating, or the like. Subsequently, a planarization process (e.g., CMP) is performed on the conductive material, thereby forming the plurality of conductive contacts **424**. Although not shown, additional dielectric layers and conductive features may be subsequently formed over the ILD layer **422**. For example, one or more additional ILD layers, conductive wires (e.g., copper wires), conductive vias (e.g., copper vias), and/or passivation layers may be formed over the ILD layer **422**.

(71) As shown in FIG. **16**, a wafer dicing process is performed to singulate individual ICs from the SOI wafer **100**. In some embodiments, the wafer dicing process comprises performing a series of cuts into the SOI wafer **100** and the ILD layer **422** to form a plurality of scribe lines **1602**. Subsequently, a mechanical force is applied to the SOI wafer **100** to singulate the individual ICs from the SOI wafer **100**. In further embodiments, the cuts may be performed by, for example, mechanical sawing, laser cutting, or the like. It will be appreciated that, in some embodiments, the IC **400** (see, e.g., FIG. **4** or **5**) may be one of the individual ICs singulated from the SOI wafer **100**.

(72) FIG. **17** illustrates a flowchart of some embodiments of a method for forming a semiconductor-on-insulator (SOI) wafer having a composite insulator layer and singulating individual integrated chips (ICs) from the SOI wafer. While the flowchart **1700** of FIG. **17** is illustrated and described herein as a series of acts or events, it will be appreciated that the illustrated ordering of such acts or events is not to be interpreted in a limiting sense. For example, some acts may occur in different orders and/or concurrently with other acts or events apart from those illustrated and/or described herein. Further, not all illustrated acts may be required to implement one or more aspects or embodiments of the description herein, and one or more of the acts depicted herein may be carried out in one or more separate acts and/or phases.

(73) At act **1702**, a first insulator layer is formed over a handle wafer via a thermal oxidation process. FIG. **6** illustrates a cross-sectional view of some embodiments corresponding to act **1702**.

(74) At act **1704**, an etch stop layer is formed over a donor wafer. FIG. **7** illustrates a cross-sectional view of some embodiments corresponding to act **1704**.

(75) At act **1706**, a device layer is formed over the etch stop layer. FIG. **7** illustrates a cross-

sectional view of some embodiments corresponding to act **1706**.
(76) At act **1708**, a second insulator layer is formed over the device layer via a chemical vapor deposition (CVD) process. FIG. **8** illustrates a cross-sectional view of some embodiments corresponding to act **1708**.
(77) At act **1710**, the second insulator layer is bonded to the first insulator layer to form a composite insulator layer between the donor wafer and the handle wafer. FIGS. **9-10** illustrate a series of cross-sectional views of some embodiments corresponding to act **1710**.
(78) At act **1712**, the donor wafer is removed. FIG. **11** illustrates a cross-sectional view of some embodiments corresponding to act **1712**.
(79) At act **1714**, the etch stop layer is removed to form a semiconductor-on-insulator (SOI) wafer having the composite insulator layer. FIGS. **12-13** illustrate a series of cross-sectional views of some embodiments corresponding to act **1714**. In some embodiments, a method **1715** for forming the SOI wafer comprises acts **1702**, **1704**, **1706**, **1708**, **1710**, **1712**, and **1714**.
(80) At act **1716**, one or more semiconductor devices are formed on/over the device layer. FIG. **14** illustrates a cross-sectional view of some embodiments corresponding to act **1716**.
(81) At act **1718**, an interlayer dielectric (ILD) layer and a plurality of conductive contacts are formed over the one or more semiconductor devices. FIG. **15** illustrates a cross-sectional view of some embodiments corresponding to act **1718**.
(82) At act **1720**, one or more individual integrated chips (ICs) are singulated from the SOI wafer. FIG. **16** illustrates a cross-sectional view of some embodiments corresponding to act **1720**. In some embodiments, a method **1721** for forming an integrated chip (IC) comprising a semiconductor-on-insulator (SOI) substrate having a composite insulator structure comprises acts **1716**, **1718**, and **1720**.
(83) In some embodiments, the present application provides a semiconductor wafer. The semiconductor wafer comprises a handle wafer. A first oxide layer is disposed over the handle wafer. A device layer is disposed over the first oxide layer. A second oxide layer is disposed between the first oxide layer and the device layer, wherein the first oxide layer has a first etch rate for an etch process and the second oxide layer has a second etch rate for the etch process, and wherein the second etch rate is greater than the first etch rate.
(84) In some embodiments, the present application provides an integrated chip (IC). The IC comprises a semiconductor wafer. The semiconductor wafer comprises: a handle substrate; a device substrate disposed over the handle substrate; and a composite oxide structure disposed between the handle substrate and the device substrate, wherein the composite oxide structure comprises a first oxide structure and a second oxide structure, wherein the first oxide structure has a first etch rate for a hydrofluoric etch and the second oxide structure has a second etch rate for the hydrofluoric etch, and wherein the second etch rate is greater than the first etch rate. A semiconductor device is disposed on the device substrate. An interlayer dielectric layer (ILD) is disposed over the semiconductor device.
(85) In some embodiments, the present application provides a method for forming a semiconductor wafer. The method comprises forming a first oxide layer on a handle wafer via a thermal oxidation process. An etch stop layer is formed over a donor wafer. A device layer is formed over the etch stop layer. A second oxide layer is formed on the device layer via a chemical vapor deposition (CVD) process. The first oxide layer is bonded to the second oxide layer, wherein both the first oxide layer and the second oxide layer are disposed between the device layer and the handle wafer. After the first oxide layer is bonded to the second oxide layer, the donor wafer is removed via a first etching process.
(86) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages

of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A semiconductor wafer, comprising: a handle wafer; a thermal oxide layer overlying and directly contacting the handle wafer; a device layer disposed over the thermal oxide layer; and a chemical vapor deposition (CVD) oxide layer disposed vertically between and directly contacting the thermal oxide layer and the device layer, wherein the CVD oxide layer has a dielectric strength that is less than a dielectric strength of the thermal oxide layer, and wherein the CVD oxide layer has internal stress that is tensile, whereas the thermal oxide layer has internal stress that is compressive.
2. The semiconductor wafer according to claim 1, wherein the thermal oxide layer and the CVD oxide layer share a common set of elements, including silicon and oxygen.
3. The semiconductor wafer according to claim 1, wherein the thermal oxide layer extends in a closed path to completely surround the handle wafer.
4. The semiconductor wafer according to claim 1, wherein the device layer and the CVD oxide layer share a common width less than a width of the thermal oxide layer.
5. The semiconductor wafer according to claim 1, wherein the CVD oxide layer is high-density plasma CVD (HDPCVD) oxide.
6. The semiconductor wafer according to claim 1, wherein the thermal oxide layer and the CVD oxide layer have different etch rates for an etchant comprising hydrofluoric acid.
7. The semiconductor wafer according to claim 6, wherein the thermal oxide layer has a first etch rate for the etchant, wherein the CVD oxide layer has a second etch rate for the etchant, and wherein the second etch rate is greater than the first etch rate.
8. An integrated chip (IC), comprising: a semiconductor substrate comprising: a handle substrate; a device substrate disposed over the handle substrate; a composite oxide structure disposed between the handle substrate and the device substrate, wherein the composite oxide structure comprises a first oxide structure and a second oxide structure both overlying the handle substrate, wherein the second oxide structure is disposed vertically between the first oxide structure and the device substrate, and wherein the second oxide structure has a greater density than the first oxide structure; and a third oxide structure underlying and directly contacting the handle substrate, wherein the third oxide structure has an etch rate for hydrofluoric acid that is less than an etch rate that the second oxide structure has for hydrofluoric acid; a semiconductor device disposed on the device substrate; and an interlayer dielectric (ILD) layer disposed over the semiconductor device.
9. The IC according to claim 8, wherein the first and second oxide structures have different intrinsic stresses.
10. The IC according to claim 8, wherein the third oxide structure shares a common width with the composite oxide structure and has a lesser thickness than the composite oxide structure.
11. The IC according to claim 8, wherein the greater density is about 2.1 to 2.3 grams per cubic centimeter (g/cm3).
12. The IC according to claim 8, wherein the first oxide structure and the second oxide structure respectively have a first etch rate and a second etch rate, and wherein a ratio of the second etch rate to the first etch rate is about 400:25 for an etchant.
13. An integrated chip (IC), comprising: a semiconductor substrate comprising: a handle substrate; a device substrate disposed over the handle substrate; and a composite insulator structure disposed between the handle substrate and the device substrate, wherein the composite insulator structure comprises a first insulator structure and a second insulator structure, wherein the second insulator structure is disposed vertically between the device substrate and the first insulator structure,

wherein the first insulator structure comprises oxide having a first dielectric strength, and wherein the second insulator structure comprises oxide having a second dielectric strength that is less than the first dielectric strength; and a semiconductor device inset into the device substrate.
14. The IC according to claim 13, wherein the first insulator structure and the second insulator structure comprise silicon oxide.
15. The IC according to claim 14, wherein the second insulator structure has a greater concentration of hydrogen, carbon, or chlorine compared to the first insulator structure.
16. The IC according to claim 13, wherein the second dielectric strength is less than about 11 megavolts per centimeter (MV/cm), and wherein the first dielectric strength is greater than or equal to about 11 MV/cm.
17. The IC according to claim 13, wherein the first insulator structure is thermal oxide, and wherein the second insulator structure is non-thermal oxide.
18. The IC according to claim 13, wherein the first insulator structure contacts the second insulator structure, wherein the first insulator structure has a first etch rate and the second insulator structure has a second etch rate for a common etchant, and wherein the second etch rate is greater than the first etch rate.
19. The IC according to claim 13, wherein the second insulator structure has internal stress that is tensile, whereas the first insulator structure has internal stress that is compressive.
20. The IC according to claim 13, wherein the first insulator structure has internal stress that is compressive and the second insulator structure has internal stress that is compressive.